BlackJack Program

Software Requirements Specifications

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Revision History

Date	Revision	Description	Author
02/20/2026	1.0	Initial Version	All Members
03/01/2026	1.1	Added Definitions, Overview, Constraints	Nikki Ong
03/02/2026	1.2	Added Functional and Modules requirements	Monisha Turan
03/02/2026	1.3	Added Assumptions, Scope, Product Perspective & Architecture	Tochi Ezeife
03/02/2026	2.0	Added External & Internal Requirements	Sadhvik Chelmatikary
03/02/2026	2.1	Added Non-Functional Requirements	Daniel Alvarado
03/04/2026	2.2	Edited Format/Layout & Add Class Diagram	Nikki Ong

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1. Purpose

This document outlines the requirements for a Multiplayer Black Jack Gaming System.

1. Scope

This document will detail the user, GUI, functional and non-functional requirements for the BlackJack Simulation Program (Name pending). This document will not document how these requirements will be implemented. It is a blueprint that will be closely adhered to during the design phase.

2. Definitions, Acronyms, Abbreviations

Dealer - Authorized personnel responsible for shuffling and dealing the deck of cards in accordance with Blackjack regulations

Hand - The cards held by the player/dealer during the game.

Hit - the action of acquiring a card to be dealt by the dealer to the player's hand.

Stand - the action of refusing to be dealt more cards from the dealer.

Bust - when the accumulation of cards the dealer deals exceeds 21.

Ace - in Blackjack, an Ace card can be represented as a value of 1 or 11

Natural - Player's first two cards are an Ace and a ten point card (Jack, Queen, King, or a 10 card) equaling an immediate value of 21.

3. References

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4.Overview

The Multiplayer Black Jack Gaming System, is designed to simulate a real-time casino experience, allowing multiple players to simultaneously compete against a real-time dealer. Where players aim to reach a hand value of 21 without going over that number.

2. Overall Description

1. Product Perspective

Create a multi-player blackjack gaming system. The system allows clients to log in, add funds to their accounts, play games of blackjack with other real players, or cash out their accounts. Standard rules apply. Cheating is minimized. Play is fair. This is a Java application with a GUI that operates over TCP/IP. This system requires a server application and client application. There is no web or HTML component. No databases, libraries, frameworks, or other technologies may be used without approval.

2. Product Architecture

The system will be organized into 5 major modules:

1. User module
2. Server module
3. Game Logic module
4. Networking module
5. Client/GUI module

Note: System architecture should follow standard OO design practices.

3. Product Functionality/Features

The high-level features of the system are as follows (see section 3 of this document for more detailed requirements that address these features):

- Being able to create an account and login.
- To add/withdraw funds from player's accounts.
- Allow multiple players to join and play at tables.
- Automatic table management
- Dealer Functionality, enforcing Blackjack rules.
- GUI displaying gameplay.
- Server managing game logic, player sessions, and fairness within the game.
- Communication between client application and server using network sockets.

4. Constraints

2.4.1 The system must be implemented entirely in Java.

2.4.2 Since the system has to operate as a Java application it means there mustn't have web-based code be used (HTML, CSS, JavaScript).

2.4.3 The system must use Java socket connections due to it being a client-server application.

2.4.4 The server has to handle client disconnections without stopping active games/other connections in the server system.

2.4.5 The server must support multiple client connections since multiple players can play at the same time.

2.4.6 The system has to correctly handle Blackjack standard rules such as an Ace valuation (it being value 1 or 11) & recognize a natural BlackJack on the first deal.

2.4.7 Players must not lose money unfairly, meaning their account balance can’t drop below zero where bets must be validated by the server before each round.

2.4.8 Only registered users with valid credentials can connect to the server, since players have to be authenticated before joining a game.

2.4.9 The system must not allow players to access other player’s hands (each player’s cards in game) or manipulate game outcomes as the system has to be fair.

2.4.10 The system must maintain valid decks for each table.

5. Assumptions and Dependencies

2.5.1 It is assumed that the maximum # of players at a table active at any time will be 35

2.5.2 It is assumed that the system will **not** be running at all times.

2.5.3 It is assumed that any number of players can login at any time.

2.5.4 It is assumed that all users have access to the internet.

2.5.5 It is assumed that one dealer will be assigned to an active table.

3. Specific Requirements

3.1 Functional Requirements

3.1.1 Common Requirements:

3.1.1.1 The system shall allow registered users to log in using a valid username and password.

3.1.1.2 The system shall allow multiple players to connect and play simultaneously.

3.1.1.3 Each table shall contain one Dealer and up to six Player users.

3.1.1.4 Users shall be required to place a bet before cards are dealt.

3.1.1.5 The system shall automatically calculate hand totals.

3.1.1.6 The system shall determine win, loss, push, or blackjack outcomes at the end of each round.

3.1.1.7 The system shall display real-time game updates to all players at the table.

3.1.1.8 The system shall prevent player account balances from dropping below zero.

3.1.1.9 A new table shall be created when all seats at the current table are occupied.

3.1.1.10 Players that do not perform an action within 90 seconds will automatically fold

3.1.1.11 Players shall have the ability to replay and join the next round after a game concludes.

3.1.2. User Module Requirements:

3.1.2.1 allows users to create an account with a unique username and password (6–20 characters).

3.1.2.2allows users to log in using their registered credentials.

3.1.2.3 allows users to deposit money into their account before joining a game.

3.1.2.4 The system shall allow users to cash out their available balance.

3.1.2.5 displays the user's current balance on the screen.

3.1.2.6 prevents users from placing bets higher than their available balance.

3.1.2.7 The system shall allow users to join an available table.

3.1.2.8 The system shall allow users to choose to hit, stand, or fold during their turn.

3.1.3. Server Module Requirements:

3.1.3.1 Handle multiple players connected at the same time.

3.1.3.2 Verify the login information before allowing access.

3.1.3.3 Server shall shuffle and deal all cards.

3.1.3.4 Control turn order in clockwise direction.

3.1.3.5 Calculate hands total and determine winners.

3.1.3.6 Update player balances after each round.

3.1.3.7 Continue running even if a player disconnects.

3.1.4. Game Logic Module Requirements:

3.1.4.1 Follow standard Blackjack rules during the game.

3.1.4.2 Deals two cards to each player and the dealer at the start of a round.

3.1.4.3 The system shall determine when a player busts (goes over 21).

3.1.4.4 The dealer shall stand when their hand reaches 17 or higher.

3.1.4.5 The system shall treat an Ace as either 1 or 11.

3.1.4.6 Reset the game state after each round ends.

3.1.5. Networking Module Requirements:

3.1.5.1 The client shall connect to the server using Java TCP/IP sockets.

3.1.5.2 The system shall send player actions from the client to the server.

3.1.5.3 The server shall send updated game information back to all players.

3.1.5.4 Keeps all players' game view synchronized.

3.1.6. Client/GUI Module Requirements:

3.1.6.1 Provides interactive controls for player actions (Hit, Stand, Fold)

3.1.6.2 Sends player actions to the server

3.1.6.3 Displays the player’s current hand of cards

3.1.6.4 Displays the dealer’s one card faced up and other card faced down

3.1.6.5 Displays the player’s account balance and current bet

3.1.6.6 Displays game status messages

3.1.6.7 Displays the results of rounds (Win, Lost, Bust)

3.1.6.8 Client/GUI updates based on server responses

3.2. External Interface Requirements

3.2.1 The system must provide an interface to the User module so that the users can see their dealt cards and make decisions in the game. The interface should be of a visual format with the following fields: cards, current balance

3.2.2 The system must provide an interface to the User module so they can login to the system. The interface should consist of 2 text fields: username and password. The user must be able to provide input in the form of text. The system must also give a visual cue if there was an error in any of the entries. The system should suggest the user to try inputting their information again.

3.2.3 The system must provide an interface to end their game session through a visual cue.

3.2.4 The system must provide an interface across different computers to share their game data and progress over TCP protocol.

3.3 Internal Interface Requirements

3.3.1 The system must represent each card with a different number value. The system must use the card's value to compute score
3.3.2 The system must randomize the card of decks with built in pseudorandom generator before shuffling each time
3.3.3 The system should fill the table with different players based on the queue of waiting clients in a first come first serve fashion.
3.3.4 The system must be able to restrict the number of cards to be drawn after the amount 21 has been exceeded by the users

4. Non-Functional Requirements

1. Security and Privacy Requirements X

4.1.1 User logins are password protected, users can only access an account with a valid username & password matching our records.

4.1.2 Passwords can consist of the following: (char: A-Z, int: 0-9, symbols:_!?@#&) and are case sensitive.

4.1.3 Username & Password details will be saved via an ordered list.

2. Environmental Requirements

4.2.1 system must be running windows 10 or higher, mac os 13 or linux
4.2.2 system must be able to support sdk java 17
4.2.3 system must have a minimum of 720p resolution
4.2.4 system must have keyboard/mouse either built in or plug in
4.2.5 system must have wifi capabilities and be able to connect to a network
4.2.6 system must have cpu of atleast intel i5 or amd ryzen 5
4.2.7 system must have at least 8gb RAM and a graphics card that is able to render a GUI.

3. Performance Requirements

4.3.1 system will process players' action inputs within 1 second.
4.3.2 system will handle an infinite number concurrent players and dealers.

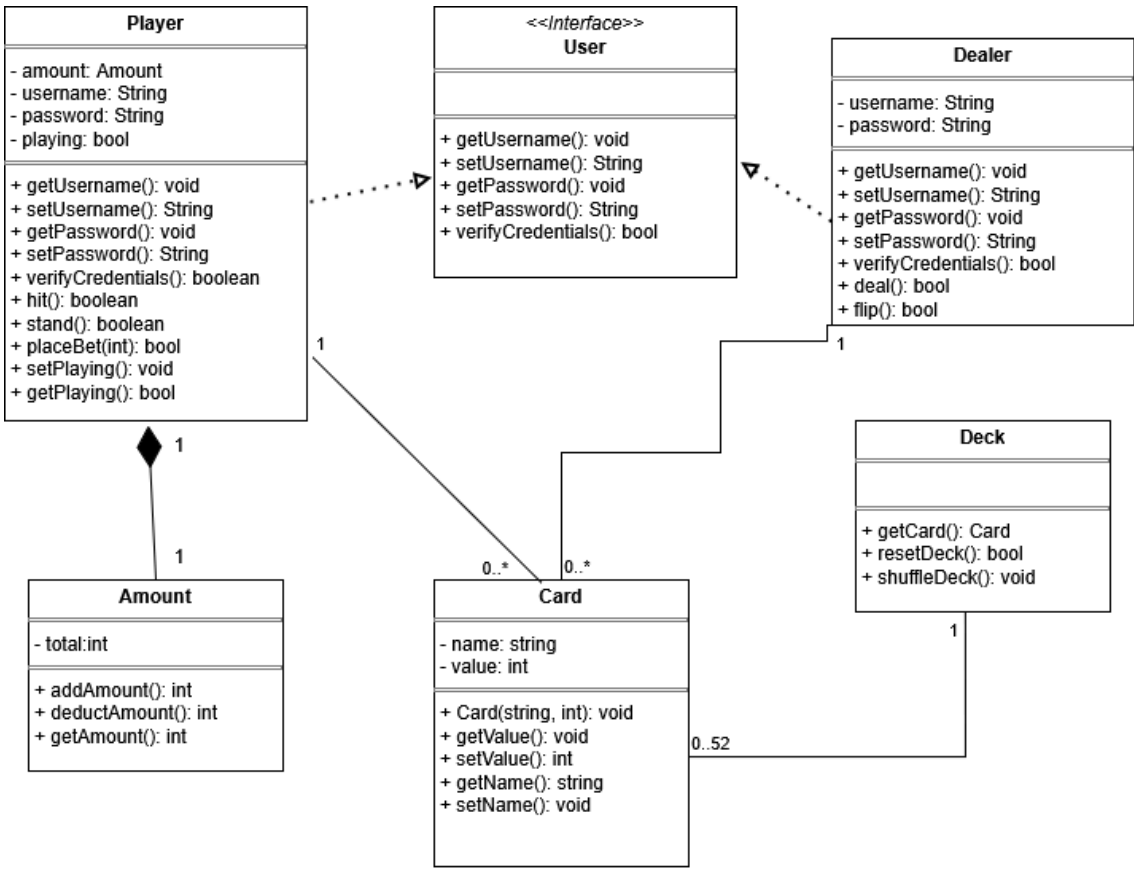
Use Case Specifications:

(insert use cases when all done, formatted wtv)

UML Use Case Diagrams Document:

(insert use case UML here)

Class Diagram:



Sequence Diagrams:

- Player Joins Table

(insert diagram here)

- Player Action: “Hit”